

A FINITE PRIME-SET EXCLUSION AND AN ODD-MODULAR OBSTRUCTION IN LEHMER'S TOTIENT PROBLEM

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ABSTRACT. Lehmer's totient problem asks whether a composite integer n can satisfy $\varphi(n) \mid n - 1$. The problem remains open. We prove the certified finite result that every composite solution, if one exists, has largest prime factor at least 349. Two independently written exact prime-set enumerations leave 38 sets after the 2-adic filter; a single local congruence modulo 3 rejects all of them. We also prove an infinite obstruction theorem: compatibility, reciprocal mass, and the complete 2-adic divisibility condition do not imply the odd part of the Korselt condition. Thus odd modular information is indispensable. These are bounded and structural results, not a resolution of Lehmer's problem or an improvement of the known global bounds on the size or number of prime factors.

1. INTRODUCTION

For a positive integer n , let $\varphi(n)$ denote Euler's totient function. In 1932 Lehmer asked whether

$$(1.1) \quad \varphi(n) \mid n - 1$$

can hold for a composite n [6]. No such integer is known, and the problem is still open [10]. The strongest globally cited computational restrictions located in our literature audit are

$$\omega(n) \geq 15, \quad n > 10^{30},$$

where $\omega(n)$ is the number of distinct prime divisors. The first bound was already established in Wong's 1997 thesis; Renze and Pinch later reported further independent computations, and Pinch proved the second bound [11, 9, 8]. Earlier work of Cohen and Hagi gave $\omega(n) \geq 14$ and $n > 10^{20}$ [4].

These global results imply the explicit comparison bound $P^+(n) \geq 79$. Indeed, a hypothetical solution is odd and square-free, whereas the product of all odd primes at most 73 is

$$20364840299624512075310661735 < 10^{30}.$$

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Under the additional condition that the Lehmer index $(n-1)/\varphi(n)$ is at least 3, the known bound $\omega(n) \geq 1991$ gives the much stronger conditional consequence $P^+(n) \geq 17327$ [5]. The globally difficult index-two branch is therefore the relevant one for the endpoint below.

The purpose of this note is narrower. If $P^+(n)$ denotes the largest prime divisor of $n > 1$, we certify the following bounded exclusion.

Theorem 1.1. *If n is composite and satisfies (1.1), then*

$$P^+(n) \geq 349.$$

Equivalently, there is no solution whose prime divisors all belong to the set of 68 odd primes not exceeding 347. This endpoint is modest. We make no claim that prime-set search, compatibility pruning, or product bounds are new: all have clear precedents in the computations of Wong, Renze, Pinch, and Burcsi–Czirbusz–Farkas [9, 8, 3, 11]. The contribution here is a pair of independent exact enumerations, the uniform explanation of the 38 retained sets, and the elementary refinement in Section 4.

In a dated search of the databases, sources, and public computational archives recorded with the certificate, we did not locate an earlier explicit exclusion of all candidates supported on primes at most 347. This is a bounded negative literature search, not an unqualified claim of first discovery.

2. STRUCTURAL REDUCTION

Suppose throughout this section that $n > 1$ is composite and satisfies (1.1). Put

$$k = \frac{n-1}{\varphi(n)}.$$

Lemma 2.1. *The integer n is odd and square-free. Moreover,*

$$\gcd(n, \varphi(n)) = \gcd(n, k) = 1.$$

If p and q are distinct prime divisors of n , then

$$(2.1) \quad p \nmid q-1.$$

Finally, n is a Carmichael number.

Proof. If n were even, then $n > 2$, so $\varphi(n)$ would be even whereas $n-1$ would be odd. If $p^2 \mid n$, then $p \mid \varphi(n)$, and (1.1) would imply $p \mid n-1$, contradicting $p \mid n$. Thus n is odd and square-free.

Every common divisor of n and $\varphi(n)$, or of n and k , divides both n and $k\varphi(n) = n-1$; both greatest common divisors are therefore one. If $p \mid q-1$ with $p, q \mid n$, then $p \mid \varphi(n)$, contrary to $\gcd(n, \varphi(n)) = 1$. Finally, for every $p \mid n$, $p-1 \mid \varphi(n) \mid n-1$. Fermat's theorem modulo each p , followed by the Chinese remainder theorem, gives $a^n \equiv a \pmod{n}$ for every integer a . Hence n is Carmichael. $\square$

Write

$$n = \prod_{p \in S} p, \quad \Phi(S) = \varphi(n) = \prod_{p \in S} (p-1), \quad R(S) = \prod_{p \in S} \frac{p}{p-1}.$$

The compatibility graph has the odd primes as vertices; for $p < q$, the pair is compatible exactly when $p \nmid q-1$. Lemma 2.1 says that the prime set of a solution is a clique in this graph.

Lemma 2.2. *For a composite solution,*

$$(2.2) \quad R(S) = k + \frac{1}{\Phi(S)} > 2.$$

If C is a set of compatible undecided primes at a search node, then every completion $S \cup T$, $T \subseteq C$, satisfies

$$(2.3) \quad R(S \cup T) \leq R(S \cup C).$$

Consequently, a node for which $R(S \cup C) \leq 2$ contains no solution.

Proof. Dividing $n = k\Phi(S) + 1$ by $\Phi(S)$ gives the equality in (2.2). Since n is composite, $\varphi(n) \neq n-1$, so $k \geq 2$. Every factor $p/(p-1)$ exceeds one, which proves (2.3) and the pruning statement. $\square$

The Lehmer equation also has a useful local form.

Lemma 2.3. *For every $p \in S$,*

$$(2.4) \quad k \prod_{\substack{q \in S \\ q \neq p}} (q-1) \equiv 1 \pmod{p}.$$

In particular, if $3 \in S$, then $k \equiv 1 \pmod{3}$.

Proof. Reducing $n = k\Phi(S) + 1$ modulo p and using $p-1 \equiv -1 \pmod{p}$ gives (2.4). If $3 \in S$, compatibility excludes $q \equiv 1 \pmod{3}$ for every other $q \in S$. Hence $q \equiv 2 \pmod{3}$, every factor $q-1$ in (2.4) is one modulo 3, and $k \equiv 1 \pmod{3}$. $\square$

The pruning comparison is evaluated without division: if $R(S \cup C) = A/B$ is kept as the unreduced product of integer numerators and denominators, the node is rejected exactly when $A \leq 2B$.

3. THE EXACT FINITE SEARCH

Let

$$\mathcal{P}_{347} = \{p : p \text{ is an odd prime and } p \leq 347\}.$$

This set has 68 elements, the largest being 347. The search maintains a selected compatible set S , a set C of compatible undecided primes, the exact products $n(S)$, $\Phi(S)$, and the modulus

$$L(S) = \text{lcm}_{p \in S} (p-1).$$

At a node, the least prime $p \in C$ is selected in one branch and omitted in the other. The selection branch removes every later q for which $p \mid q-1$.

TABLE 1. Exact search counts for the prime cap 347.

Quantity	Count
Odd primes in the universe	68
Compatible unordered pairs	2,183
Incompatible unordered pairs	95
Canonical selection attempts	27,179,711
Subtrees removed by the exact ratio bound	5,457,679
Nodes surviving the ratio upper bound	21,722,032
Sets with $R(S) > 2$	17,803,010
Sets passing the 2-adic condition	38
Sets passing the Korselt lcm condition	0
Sets passing the full Lehmer divisibility	0

Before descending, Lemma 2.2 is applied to the entire compatible tail by integer cross-multiplication.

Every set with $R(S) > 2$ is tested against the following necessary conditions, in order:

$$(3.1) \quad 2^{v_2(\Phi(S))} \mid n(S) - 1,$$

$$(3.2) \quad L(S) \mid n(S) - 1,$$

$$(3.3) \quad \Phi(S) \mid n(S) - 1.$$

Condition (3.1) is one prime-power component of (3.3). Condition (3.2) collects the Carmichael congruences $p - 1 \mid n - 1$. Both are rejection filters only; neither is treated as sufficient for the Lehmer condition.

Proposition 3.1. *The canonical search over $\mathcal{P}_{347}$ has the exact counts in Table 1. In particular, 38 sets pass (3.1), no set passes (3.2), and hence no set passes (3.3). Every one of the 38 sets contains 3 and satisfies*

$$(3.4) \quad 2 < \frac{n(S)}{\Phi(S)} < 3.$$

Proof. The search program is part of the certificate. Its state is a bit set for C , a bit set for S , and exact Python integers for all products and moduli. Sieve generation gives the 68 odd primes at most 347. Compatibility is tested by the exact remainder $(q - 1) \bmod p$.

Completeness follows by induction on C . The select/omit branches partition all subsets containing or not containing its least element. In the select branch, only primes forbidden by (2.1) are removed. The only whole-subtree numerical rejection is $R(S \cup C) \leq 2$, which is valid by Lemma 2.2. Thus every compatible subset capable of satisfying (2.2) is visited exactly once. Equations (3.1)–(3.3) are evaluated by integer remainders.

The serialized record contains all 38 sets that pass (3.1), together with $n(S)$, $\Phi(S)$, $v_2(\Phi(S))$, $L(S)$, and the nonzero value of $(n(S) - 1) \bmod L(S)$.

Hence every one of the 38 sets fails (3.2). Exact cross multiplication also verifies (3.4) and direct membership testing verifies that 3 occurs in every retained set. If such a set satisfied the Lehmer equation, (3.4) would force $k = 2$, whereas Lemma 2.3 would force $k \equiv 1 \pmod{3}$. Thus the entire 38-element residue can alternatively be rejected by one local congruence. The counts and the complete retained list are reproduced by the independent verifier described in Section 5. $\square$

Proof of Theorem 1.1. By Lemma 2.1, a composite solution with $P^+(n) \leq 347$ would give a compatible subset of $\mathcal{P}_{347}$. Lemma 2.2 and Proposition 3.1 exclude every such subset. The next prime after 347 is 349. Therefore every composite solution has $P^+(n) \geq 349$. $\square$

4. AN ELEMENTARY ODD-OBSTRUCTION REFINEMENT

The 38 survivors suggest asking whether compatibility, reciprocal mass, and the complete 2-adic condition might themselves force the odd part of the Korselt condition. The following theorem rules out any such implication, and does so at an arbitrarily prescribed fresh odd prime power. The construction refines the standard Dirichlet extension of a finite normal family already used by Wong. Related but different infinitude results appear in work on Korselt sets and radimichael numbers [11, 1, 7]. We make no absolute priority claim for the refinement.

Theorem 4.1. *Let B be a finite compatible set of odd primes, put $m = \prod_{p \in B} p$ and $A = \prod_{p \in B} (p - 1)$, and suppose $m > 2A$. Let ℓ be an odd prime for which*

$$\ell \nmid mA(m - 1),$$

and let $b \geq 1$. There are infinitely many primes $q > \max B$ such that $S = B \cup \{q\}$ is compatible and

$$\frac{n(S)}{\Phi(S)} > 2, \quad 2^{v_2(\Phi(S))} \mid n(S) - 1,$$

but $\ell^b \mid L(S)$ while $\ell \nmid n(S) - 1$.

Proof. Set $a = v_2(m - 1)$, $V = v_2(A)$, and $E = V + a$. The inequality $m > 2A$ makes B nonempty, so $V \geq 1$ and $E > a$. By the Chinese remainder theorem, impose

$$(5.1) \quad \begin{aligned} q &\equiv m^{-1} \pmod{2^E}, \\ q &\equiv 2 \pmod{p} \quad (p \in B), \\ q &\equiv 1 \pmod{\ell^b}. \end{aligned}$$

The moduli are pairwise coprime, and the resulting residue class is coprime to their product. Dirichlet's theorem on primes in arithmetic progressions therefore supplies infinitely many prime values of q [2]; discard the finitely many with $q \leq \max B$.

The congruences $q \equiv 2 \pmod{p}$ show that no $p \in B$ divides $q - 1$, while $q > p$ gives $q \nmid p - 1$; compatibility is preserved. Also

$$\frac{n(S)}{\Phi(S)} = \frac{m}{A} \frac{q}{q-1} > 2.$$

The first congruence in (5.1) gives

$$m(q-1) \equiv -(m-1) \pmod{2^E},$$

and hence $v_2(q-1) = a$. Thus $v_2(\Phi(S)) = V + a = E$, while the same congruence gives $mq \equiv 1 \pmod{2^E}$. This proves the complete 2-adic divisibility. Finally, $\ell^b \mid q-1$ puts ℓ^b into $L(S)$, whereas

$$mq - 1 \equiv m - 1 \not\equiv 0 \pmod{\ell}.$$

□

For example, the base $B = \{3, 5, 17, 23\}$, $\ell = 7$, and $b = 1$ gives the prime $q = 129493337$. Exact arithmetic verifies every conclusion of Theorem 4.1. The theorem does not exclude all potential Lehmer supports. Its role is sharper: it proves that an odd modular reachability condition cannot be inferred from the inexpensive compatibility, ratio, and 2-adic filters alone.

5. INDEPENDENT VERIFICATION AND BOUNDARY AUDIT

The replay program was written separately from the search program. It generates primes by deterministic trial division rather than a sieve, stores candidate states as prime tuples rather than graph bit sets, and rebuilds the exact upper product at each branch. It reproduces all deterministic counts, all 38 retained tuples, their exact arithmetic data, and the empty final solution list. Three additional tests check:

- (1) agreement of the two prime generators through 347;
- (2) agreement with an unpruned full power-set enumeration through 30;
- (3) agreement of the two branch-and-bound implementations through 50.

All tests pass.

The boundary cases are handled before or inside the enumeration. The integer 1 is outside the composite problem. A prime is not a counterexample. Even and non-square-free composites are excluded analytically in Lemma 2.1. The empty and one-element sets cannot represent a composite solution. The ratio cutoff uses a non-strict comparison at its boundary, while an actual solution has the strict inequality (2.2). Repeated prime powers in the factors $p-1$ are retained in $\Phi(S)$; the lcm test is used only as a weaker preliminary filter.

The theorem does not assert that the endpoint 349 is attained. It says only that every hypothetical solution lies beyond the exhausted prime universe.

6. REPRODUCIBILITY, PRIORITY, AND LIMITATIONS

The certificate archive contains the exact search, an independent verifier, unit and comparison tests, and the serialized search record. The file `certificates/README.md` identifies every component and states the trust boundary. From the project directory, the complete certificate is checked by

```
tests/run_full_certificate.sh
```

The computation uses Python integers; floating-point values are used only for initial guesses that are corrected by integer-power comparisons, and for elapsed-time reporting. They never decide a mathematical inequality. A proof assistant was not used.

The main limitation is mathematical, not computational: the result says nothing about a prime set containing a prime greater than 347. It does not prove Lehmer's conjecture, produce a counterexample, or improve the published global bounds $\omega(n) \geq 15$ and $n > 10^{30}$. Under the additional hypothesis $3 \mid n$, much stronger conditional bounds are already known [3]; under larger Lehmer index, earlier work also gives much larger conditional bounds [5]. Theorem 4.1 also shows why simply iterating the ratio and 2-adic filters cannot produce an unbounded argument: odd modular information is independent of them.

The accompanying priority audit was completed on 22 August 2026. It covered the recorded MR Lookup entries, zbMATH title and citation-chain searches, original papers, Wong's thesis, Renze's notebook, Pinch's poster and public tables, Guy's problem list, OEIS links, and modern Carmichael-number tables. No earlier explicit cap-347 exclusion was located. Full MathSciNet topic search required an unavailable institutional login, and unindexed private computations cannot be ruled out. Accordingly, the numerical result is stated as a certified bounded exclusion, without an absolute priority claim. A substantive next step would use odd modular reachability to prove an unrestricted structural bound, rather than merely raising the cap.

COMPUTATIONAL AND AI ASSISTANCE DISCLOSURE

OpenAI Codex was used for literature triage, proof exploration, code drafting, adversarial review, and language editing. The mathematical statements and computational claims were checked against primary sources and by independent exact-arithmetic implementations. Responsibility for the final manuscript and its claims rests with the author.

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